Sida Ye

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WORK EXPERIENCES

Harvard Medical School/Brigham and Women's Hospital/HHMI — Nov 2025 - Present
Postdoctoral scientist (Advisor: Prof. Matthew K. Waldor) — Boston, U.S.

University of Massachusetts Boston — June 2025 - Oct 2025
Postdoctoral fellow (Advisors: Prof. Kourosh Zarringhalam & Prof. Manoj T. Duraisingh) — Boston, U.S.

EDUCATION

University of Massachusetts Boston/Harvard T.H. Chan School of Public Health — Sep 2021- May 2025
Ph.D. in Computational Sciences — Boston, U.S.
Thesis: ANALYTICAL APPROACHES FOR IDENTIFICATION OF ESSENTIAL GENOMES OF PLASMODIUM KNOWLESI AND BABESIA DIVERGENS
Advisors: Prof. Kourosh Zarringhalam & Prof. Manoj T. Duraisingh

Peking University, School of Life Sciences — 2014 - 2020
M.S., Botany and Molecular Biology — Beijing, China

University of Cambridge, Sainsbury Laboratory — Sep 2017 - Dec 2017
Visiting student — Cambridge, U.K.

Huazhong Agricultural University, Zhidong Zhang Class — 2010 - 2014
B.S., Plant sciences and technology — Wuhan, China

FELLOWSHIPS, HONORS & AWARDS

Oracle Doctoral Research Fellowship	2025
Founder Scholarship (Peking University)	2019
First-class Scholarship for Graduate Students (Peking University)	2017
Lv Yichang Scholarship (Peking University)	2015
Outstanding students scholarship (Huazhong Agricultural University)	2012-2014
National Encouragement Scholarship (Ministry of Education in China)	2011

PUBLICATIONS

Note: # denotes co-first authorship. Asterisks and boldface indicate the representative publications/manuscripts.

1. * Elsworth B#, **Ye S#,** Dass S#, Tennessen JA, Sultana Q, Thommen BT, Paul AS, Kanjee U, Gruring C, Ferreira MU, Gubbels MJ, Zarringhalam K, Duraisingh MT. The essential genome of Plasmodium knowlesi reveals determinants of antimalarial susceptibility. **Science**. 2025;387(6734):eadq6241. PMID: 39913579; PMCID: PMC12104972; DOI: 10.1126/science.adq6241.
2. Keroack CD, Elsworth B, Tennessen JA, Paul AS, Hua R, Ramirez-Ramirez L, **Ye S**, Moreira CK, Meyers MJ, Zarringhalam K, Duraisingh MT. Comparative chemical genomics in Babesia species identifies the alkaline phosphatase PhoD as a determinant of antiparasitic resistance. **Proceedings of the National Academy of Sciences of the United States of America**. 2024;121(9):e2312987121. PMID: 38377214; PMCID: PMC10907312; DOI: 10.1073/pnas.2312987121.
3. Zarringhalam K, **Ye S**, Lou J, Rezvani Y, Gubbels MJ. Cell cycle-regulated ApiAP2s and parasite development: the Toxoplasma paradigm. **Current Opinion in Microbiology**. 2023;76:102383. PMID: 37898053; PMCID: PMC10840917; DOI: 10.1016/j.mib.2023.102383.
4. Wang DH, Liu N, **Ye S**, Chen ZS, Lin YN, Liu ZH, Xu ZH, Bai SN. Rice tapetum differentiation is sensitive to downregulation of OsUCH3, a ubiquitin C-terminal hydrolase. **Plant Biotechnology Journal**. 2023;21(7):1314-1316. PMID: 37057895; PMCID: PMC10281597; DOI: 10.1111/pbi.14052.
5. Wei G, Li S, **Ye S**, Wang Z, Zarringhalam K, He J, Wang W, Shao Z. High-resolution small RNAs landscape

provides insights into alkane adaptation in the marine alkane-degrader Alcanivorax dieselolei B-5. **International Journal of Molecular Sciences**. 2022;23(24):15995. PMID: 36555635; PMCID: PMC9788540; DOI: 10.3390/ijms232415995.
6. Zheng Y, Wang D, **Ye S**, Chen W, Li G, Xu Z, Bai S, Zhao F. Auxin guides germ-cell specification in Arabidopsis anthers. **Proceedings of the National Academy of Sciences of the United States of America**. 2021;118(22):e2101492118. PMID: 34031248; DOI: 10.1073/pnas.2101492118.

Manuscripts in preparation / about to be submitted
7. * **Ye S#**, Dass S#, Elsworth B#, Paul AS, Zarringhalam K, Duraisingh MT. Uncovering the essentialome of Babesia divergens by piggyBac transposon mutagenesis. Manuscript in preparation/about to be submitted.
8. Kumar M#, **Ye S#**, Diffendall G#, Zarringhalam K, Duraisingh MT. Lactylation of epigenetic regulators controls gene expression linked to sexual commitment in Plasmodium falciparum. Manuscript in preparation/about to be submitted.
9. * Mann A#, **Ye S#**, Sullivan M, Zarringhalam K, Duraisingh MT. Inducible transposon mutagenesis in Plasmodium falciparum reveals the essential genes of IPP bypass in the apicoplast. Manuscript in preparation/about to be submitted.
10. * Zingl FG#, **Ye S#**, Sinha A, Balazs GI, Leitner D, Basta DW, Waldor MK. Spatiotemporal MultiCAST in vivo screens identify the genetic dynamics of Vibrio cholerae requirements for intestinal colonization. Manuscript in preparation/about to be submitted.

Book
11. Ye S, Hemmati M. Predicting the Unseen: AI-Driven Surveillance and Evolutionary Modeling of Antimicrobial Resistance. CRC Press volume, Artificial Intelligence in Biomedical Research: Foundations, Frontiers, and Future Directions.(Under review)

PRESENTATIONS&MEETINGS

Sida Ye et al. “The essential genome of Plasmodium knowlesi reveals determinants of antimalarial susceptibility”. Poster in HHMI Science Meeting, Oct 20-22th, 2026
Sida Ye et al. “The essential genome of Plasmodium knowlesi reveals determinants of antimalarial susceptibility”. Invited talk in the University of Hong Kong, March 17th, 2026
Sida Ye et al. “piggyBac transposon technology vs CRISPR in whole genome screens”. Invited talk in Chinese Academy of Science, March 7th, 2026
Sida Ye et al. “The transposon-based whole genome screening in Apicomplexa parasites”. Invited talk in Tongji University, March 6th, 2026
Sida Ye et al. “The essential genome of Plasmodium knowlesi and Babesia divergens reveal the conserved and divergent essential pathways”. Invited talk in Chinese Genomics Meet-up online, Oct 22th, 2025
Sida Ye et al. “Uncovering the essentialome of Babesia divergens by piggyBac transposon mutagenesis”. Poster presented at: MOLECULAR PARASITOLOGY MEETING, Sep 14-18, 2023
Sida Ye et al. “Single-cell transcriptomic analysis of cold-induced quiescence in Babesia divergens”. Poster presented at: MOLECULAR PARASITOLOGY MEETING, Sep 14-18, 2022

PATENTS

1. Wang DH, Bai SN, **Ye S**, Xu ZH. Rice stamen-specific expression promoter and application thereof. Chinese patent CN107435044B; application CN201610352796.6A. Assignee: Peking University. Filed May 25, 2016; published Dec 5, 2017; granted Nov 20, 2020. Google Patents.
2. Wang DH, Bai SN, **Ye S**, Xu ZH. Plant pistil-specific promoter and application thereof. Chinese patent CN107760679B; application CN201610697199.7A. Assignee: Peking University. Filed Aug 19, 2016; published Mar 6, 2018; granted Dec 25, 2020. Google Patents.
3. Wang DH, **Ye S**, Bai SN, Xu ZH. Rice stamen-specific expression promoter pSSP2 and application thereof. Chinese patent CN109913448B; application CN201711327507.8A. Assignee: Peking University. Filed Dec 13, 2017; published Jun 21, 2019; granted Aug 4, 2020. Google Patents.
4. Wang DH, **Ye S**, Bai SN, Xu ZH. Rice stamen-specific expression promoter pSSP3 and application thereof.

Chinese patent CN109913450B; application CN201711329615.9A. Assignee: Peking University. Filed Dec 13, 2017; published Jun 21, 2019; granted Aug 25, 2020. Google Patents.
5. Wang DH, **Ye S**, Bai SN, Xu ZH. Use of Os350 protein and its coding gene in regulating plant fertility. Chinese patent CN110117316B; application CN201910327117.3A. Assignee: Peking University. Filed Apr 23, 2019; published Aug 13, 2019; granted Aug 4, 2020. Google Patents.

TEACHING & MENTORING EXPERIENCES

Research Mentor, University of Massachusetts
Qudseen Sultana, Master student in Computer Science 2022-2024
Current position: computer science teacher at Arlington High School and the Ottoson Middle School in Arlington, Massachusetts

Prarthana Prathap, Master student in Computer Science 2022-2024
Current position: Software Engineer in Google

Guest lecturer, University of Massachusetts 2024
INTR-D 601
Lecture on mutagenesis technology in pathogens with students in 40min

Research Mentor, Peking University
Yitong Ma, Undergraduate student 2016-2018
Graduated from California Institute of Technology, Michael Elowitz lab
Current position: Postdoc in Stanford University, Stanley Qi lab

Teaching Assistant, Peking University
Molecular biology technology course 2016

SERVICE & ENGAGEMENT

Peer Reviewer
Gen² 2026 | ICLR 2026, BMC Infectious Diseases, BioMedInformatics, Microorganisms, International Journal of Molecular Sciences

Founder, SDbioinfo 2023 - Present
https://sdbioinfo.com/
Lead a volunteer team(16 people) providing bioinformatics, system biology and AI education through articles, researcher interviews, and online seminars. Reach +300, 000 followers and audiences and have organized 22 scientific seminars and podcasts.